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Silicon germanium fins and integration methods

Abstract

A structure is provided, the structure comprising a substrate and a first silicon germanium fin over the substrate. A first silicon germanium layer may be arranged in the substrate, whereby the first silicon germanium layer may be coupled to the first silicon germanium fin. A second silicon germanium layer may be arranged in the substrate, whereby the second silicon germanium layer may be coupled to the first silicon germanium fin.

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Background/Summary

FIELD OF THE INVENTION

(1) The disclosed embodiments relate generally to semiconductor device structures, and more particularly, to silicon germanium fins.

BACKGROUND

(2) Transistors are fundamental device elements of modern digital processors and memory devices and have found applications in high-power electronics. One type of transistor that has emerged within the metal oxide semiconductor field effect transistors (MOSFETs) family of transistors is a fin field effect transistor (finFET). In a finFET, a conventional planar semiconducting channel is replaced by a semiconducting fin that extends outward from the substrate surface. In such a device, the gate, which controls current flow in the fin wraps around three sides of the fin to influence current flow from three surfaces instead of one. The improved control achieved with a finFET design results in faster switching performance and reduced current leakage than is possible with a planar transistor.

(3) Silicon germanium is a commonly known semiconductor material. The electron mobility and hole mobility of silicon germanium are greater than that of silicon. It may be desirable to have finFET transistors fabricated from silicon germanium. However, the fabrication process of finFET transistors using silicon germanium is challenging as downstream processes, for example, the

formation of isolation structures, may damage the silicon germanium fins. Thereby, there is a need for an improved silicon germanium finFET structure and integration methods to overcome the challenges mentioned above.

SUMMARY

(4) In an aspect of the present disclosure, a structure is provided, the structure comprising a substrate and a first silicon germanium fin over the substrate. A first silicon germanium layer may be arranged in the substrate, whereby the first silicon germanium layer may be coupled to the first silicon germanium fin. A second silicon germanium layer may be arranged in the substrate, whereby the second silicon germanium layer may be coupled to the first silicon germanium fin.

(5) In another aspect of the present disclosure, a structure is provided, the structure comprising a substrate and a first silicon germanium fin over the substrate. The first silicon germanium fin may have a bottom surface. A first silicon germanium layer may be arranged in the substrate. The first silicon germanium layer may be coupled to the first silicon germanium fin. The bottom surface of the first silicon germanium fin may contact a portion of the first silicon germanium layer. A second silicon germanium layer may be arranged in the substrate. The second silicon germanium layer may be coupled to the first silicon germanium fin. The bottom surface of the first silicon germanium fin may contact a portion of the second silicon germanium layer. The second silicon germanium layer may be spaced from the first silicon germanium layer.

(6) In yet another aspect of the present disclosure, a method of fabricating a structure is provided, the method comprising forming a first silicon germanium fin over a substrate. A first silicon germanium layer in the substrate may be formed, whereby the first silicon germanium layer may be coupled to the first silicon germanium fin. A second silicon germanium layer in the substrate may be formed, whereby the second silicon germanium layer may be coupled to the first silicon germanium fin.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosed embodiments will be better understood from a reading of the following detailed description, taken in conjunction with the accompanying drawings:

(2) FIG. 1 shows a structure, according to an embodiment of the disclosure.

(3) FIGS. 2 to 11 show a fabrication process flow for the structure shown in FIG. 1, according to some embodiments of the disclosure.

(4) For simplicity and clarity of illustration, the drawings illustrate the general manner of construction, and certain descriptions and details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the discussion of the described embodiments of the devices. Additionally, elements in the drawings are not necessarily drawn to scale. For example, the dimensions of some of the elements in the drawings may be exaggerated relative to other elements to help improve understanding of embodiments of the devices. The same reference numerals in different drawings denote the same elements, while similar reference numerals may, but do not necessarily, denote similar elements.

DETAILED DESCRIPTION

(5) The following detailed description is exemplary in nature and is not intended to limit the devices or the application and uses of the devices. Furthermore, there is no intention to be bound by any theory presented in the preceding background of the devices or the following detailed description.

(6) FIG. 1 shows a structure **100**, according to an embodiment of the disclosure. In one embodiment, the structure **100** may be for a vertical field effect transistor. In another embodiment, the structure **100** may be for a vertical nanowire. In yet another embodiment, the structure **100** may

be for a fin field effect transistor. The structure **100** includes a substrate **110**, a silicon germanium fin **112a**, a first silicon germanium layer **116a**, and a second silicon germanium layer **116b**. The silicon germanium fin **112a** may be arranged over the substrate **110**. The silicon germanium fin **112a** may have a first side surface **102a**, a second side surface **102b** opposite to the first side surface **102a** and a bottom surface **108** between the first **102a** and second **102b** side surfaces. The silicon germanium fin **112a** may have tapered side surfaces **102a** and **102b**, resulting in a top surface narrower than the bottom surface **108**. In one embodiment, the silicon germanium fin **112a** may have the shape of a vertical nanowire. The first silicon germanium layer **116a** may be arranged in the substrate **110** and may be coupled to the silicon germanium fin **112a**. The second silicon germanium layer **116b** may be arranged in the substrate **110** and may be coupled to the silicon germanium fin **112a**. A first side **156a** of the first silicon germanium layer **116a** and a first side **158a** of the second silicon germanium layer **116b** may underlap the bottom surface **108** of the silicon germanium fin **112a**. The first side **156a** may be arranged opposite to the first side **158a**.

(7) A portion of the bottom surface **108** near to the first side surface **102a** of the silicon germanium fin **112a** may be in direct contact with a portion of the first silicon germanium layer **116a**. Another portion of the bottom surface **108** near to the second side surface **102b** of the silicon germanium fin **112a** and different from the earlier portion of the bottom surface **108** may be in direct contact with a portion of the second silicon germanium layer **116b**. The first silicon germanium layer **116a** may have a top surface **120a**. The second silicon germanium layer **116b** may have a top surface **120b**. The top surfaces **120a** and **120b** may be co-planar with the bottom surface **108** of the silicon germanium fin **112a**. The top surface **120a** may directly contact the portion of the bottom surface **108** adjacent to the first side surface **102a** of the silicon germanium fin **112a**. The top surface **120b** may directly contact the other portion of the bottom surface **108** adjacent to the second side surface **102b** of the silicon germanium fin **112a**. The substrate **110** may have a top surface **126**. The top surfaces **120a** and **120b** may be co-planar with the top surface **126** of the substrate **110**.

(8) The silicon germanium layers **116a** and **116b** may be spaced apart such that a portion **110a** of the substrate **110** may be arranged between the first silicon germanium layer **116a** and the second silicon germanium layer **116b**. In one embodiment, the portion **110a** of the substrate **110** may have a flat top surface in direct contact with the bottom surface **108** of the silicon germanium fin **112a**. For example, the middle portion of the bottom surface **108** may be in contact with the portion **110a** of the substrate **110**. The portion **110a** may be a substrate pillar. In another embodiment, the portion **110a** may fully separate the first silicon germanium layer **116a** from the second silicon germanium layer **116b**. For example, the first silicon germanium layer **116a** may be spaced from the second silicon germanium layer **116b**. In an alternative embodiment, the first silicon germanium layer **116a** may be in direct contact with the second silicon germanium layer **116b**. For example, a top corner of the first silicon germanium layer **116a** may contact a top corner of the second silicon germanium layer **116b**. The silicon germanium layers **116a** and **116b** may have a thickness range of 1 to 20 nanometers.

(9) Another silicon germanium fin **112b** may be arranged over the substrate **110**, adjacent to and spaced from the silicon germanium fin **112a**. The silicon germanium fin **112b** may have a first side surface **106a** opposite to the second side surface **102b** of the silicon germanium fin **112a** and a second side surface **106b** opposite to the first side surface **106a**. The silicon germanium fin **112b** may have tapered side surfaces, resulting in a top surface narrower than a bottom surface. The bottom surface of the silicon germanium fin **112b** may be arranged between the first **106a** and second **106b** side surfaces. The second silicon germanium layer **116b** may extend to and may be coupled to the silicon germanium fin **112b**. In one embodiment, a second side **158b** of the second silicon germanium layer **116b** may underlap the bottom surface of the silicon germanium fin **112b** near to the first side surface **106a**. A third silicon germanium layer **116c** may be coupled to the silicon germanium fin **112b** and underlap a portion of the bottom surface of the silicon germanium fin **112b** adjacent to the second side surface **106b**. The third silicon germanium layer **116c** may be

spaced from the second silicon germanium layer **116b**. A portion of the substrate **110** directly below the silicon germanium fin **112b** may separate the third silicon germanium layer **116c** from the second silicon germanium layer **116b**.

(10) The silicon germanium fin **112a** may have side portions adjacent to the first **102a** and second **102b** side surfaces and a middle portion between the side portions and directly contacting the side portions. The middle portion of the silicon germanium fin **112a** may be directly above the portion **110a** of the substrate. The side portions of the silicon germanium fin **112a** may be directly above the portions of the first **116a** and second **116b** silicon germanium layers underlapping the silicon germanium fin **112a**.

(11) The silicon germanium fin **112b** may have side portions adjacent to the first **106a** and second **106b** side surfaces and a middle portion between the side portions and directly contacting the side portions. The middle portion of the silicon germanium fin **112b** may be directly above the portion of the substrate **110** separating the second silicon germanium layer **116b** from the third silicon germanium layer **116c**. The side portions of the silicon germanium fin **112b** may be directly above the portions of the second **116b** and third **116c** silicon germanium layers underlapping the silicon germanium fin **112b**.

(12) Silicon fins **148** and **150** may be arranged over the substrate **110**. The silicon fin **148** may be spaced apart and adjacent to the silicon fin **150**. The silicon fins **148** and **150** may be made of the same material as the substrate **110**. An isolation structure **118** may be arranged in the substrate **110** between the silicon fin **150** and the silicon germanium fin **112a**. The first silicon germanium layer **116a** may extend to the isolation structure **118**. The second side **156b** of the first silicon germanium layer **116a** may directly contact and may be adjacent to the isolation structure **118**. An interlayer dielectric layer **122** may be arranged over the substrate **110**, the silicon germanium fins **112a** and **112b**, the silicon fins **148** and **150**, the first **116a**, second **116b** and third **116c** silicon germanium layers and the isolation structure **118**. In one embodiment, upper portions of the silicon germanium fins **112a** and **112b**, and upper portions of the silicon fins **148** and **150** may protrude out of a top surface of the interlayer dielectric layer **122**. Lower portions of the silicon germanium fins **112a** and **112b** may be arranged below the top surface of the interlayer dielectric layer **122**. Similarly, lower portions of the silicon fins **148** and **150** may also be arranged below the top surface of the interlayer dielectric layer **122**. Although not shown, a dielectric liner may be arranged on the side surfaces of the lower portions of the silicon germanium fins **112a** and **112b** and the silicon fins **148** and **150**. In one example, the dielectric liner may be arranged between the interlayer dielectric layer **122** and the lower portions of the silicon germanium fins **112a** and **112b**. In another example, the dielectric liner may be arranged between the interlayer dielectric layer **122** and the lower portions of the silicon fins **148** and **150**. In one embodiment, the silicon germanium fins **112a** and **112b** may be doped n-type to be p-channel field effect transistors (pFETs) and the silicon fins **148** and **150** may be doped p-type to be an n-channel field effect transistor (nFET).

(13) The silicon germanium fins **112a** and **112b** may have a germanium concentration with a range from five percent to fifty percent. The germanium concentration of the silicon germanium fins **112a** and **112b** may not be uniformly distributed throughout the thicknesses of the fins **112a** and **112b**. For example, the germanium concentration of the side portions of the silicon germanium fins **112a** and **112b** may be higher than the germanium concentration of the middle portions of the silicon germanium fins **112a** and **112b**.

(14) The substrate **110** may be made of a suitable semiconductor material, for example, silicon, silicon carbide or silicon germanium. The isolation structure **118** may be made of a suitable dielectric material, for example, silicon dioxide or silicon nitride. The interlayer dielectric layer **122** may be made of a suitable dielectric material, for example, silicon dioxide, silicon nitride or a low dielectric constant material. The term “low dielectric constant material” may refer to a dielectric material having a dielectric constant lower than 3.9.

(15) FIGS. 2 to 11 show a fabrication process flow for the structure **100** shown in FIG. 1, according

to some embodiments of the disclosure. FIG. 2 shows a partially completed structure **100** after formation of dielectric caps **132** and a plurality of silicon fins **148**, **150**, **152** and **160**, according to an embodiment of the disclosure. Referring to FIG. 2, a layer of a suitable dielectric material, for example, silicon nitride, silicon dioxide or silicon oxynitride may be deposited over a substrate **110** and patterned by a suitable photolithography process and a wet etch or dry etch process to form dielectric caps **132**. The photolithography process may include depositing a layer of photoresist over the dielectric layer followed by exposure and developing to form suitable photoresist patterns. A wet etch or dry etch process may be used to remove portions of the dielectric layer not covered by the photoresist patterns thereby leaving behind the other portions of the dielectric layer covered by the photoresist patterns to form the dielectric caps **132**. The photoresist patterns may subsequently be removed. The dielectric caps **132** may be used as a hard mask for patterning the substrate **110** to form the silicon fins **148**, **150**, **152** and **160**. A reactive ion etching (RIE) process may be used to remove portions of the substrate **110** not covered by the dielectric caps **132**, leaving behind the other portions of the substrate **110** under the dielectric caps **132** to form the silicon fins **148**, **150**, **152** and **160**. The dielectric caps **132** on top of the silicon fins **148**, **150**, **152** and **160** protect the top surfaces of the silicon fins **148**, **150**, **152** and **160** from subsequent processing.

(16) FIG. 3 shows a partially completed structure **100** after formation of a protective liner **130**, according to an embodiment of the disclosure. Referring to FIG. 3, a layer of a suitable dielectric material, for example, silicon carbide, silicon carbon nitride, silicon oxycarbonitride, silicoboron carbonitride may be deposited over the silicon fins **148**, **150**, **152** and **160**, the dielectric caps **132** and the substrate **110** to form the protective liner **130**. The protective liner **130** may be deposited by a suitable deposition process, for example, chemical vapor deposition, atomic layer deposition, physical vapor deposition, or any other suitable deposition processes.

(17) FIG. 4 shows a partially completed structure **100** after patterning the protective liner **130**, according to an embodiment of the disclosure. Referring to FIG. 4, the protective liner **130** may be patterned by a suitable photolithography process followed by a wet etch or dry etch process. The patterning process may remove a portion of the protective liner **130** from the silicon fins **152** and **160**, the dielectric caps **132** on top of the silicon fins **152** and **160** and top surfaces of the substrate **110** adjacent to the silicon fins **152** and **160**. The patterning process may leave behind another portion of the protective liner **130** over the silicon fins **148** and **150**, the dielectric caps on top of the silicon fins **148** and **150** and top surfaces of the substrate **110** adjacent to the silicon fins **148** and **150**. The protective liner **130** and the dielectric caps **132** may be made of dielectric materials with different etch selectivity. The term “etch selectivity” may refer to a ratio of etching rates between different materials. For example, the protective liner **130** may be made of silicon carbide, silicon carbon nitride, silicon oxycarbonitride, silicoboron carbonitride while the dielectric caps **132** may be made of silicon nitride, silicon oxynitride or silicon dioxide. The protective liner **130** may be made of a material with a higher etching rate as compared to the dielectric caps **132**.

(18) FIG. 5 shows a partially completed structure **100** after formation of a silicon germanium condensation layer **136**, according to an embodiment of the disclosure. Referring to FIG. 5, the silicon germanium condensation layer **136** may be formed over the silicon fins **152** and **160**. The silicon germanium condensation layer **136** may also be formed over the top surfaces of the substrate **110** adjacent to a first side surface **102a** and a second side surface **102b** of the silicon fin **152** and top surfaces of the substrate **110** adjacent to a first side surface **106a** and a second side surface **106b** of the silicon fin **160**. The formation of the silicon germanium condensation layer **136** may include selective growth of a layer of silicon germanium by a suitable process, for example, epitaxial growth. The silicon germanium layer does not grow on the protective liner **130** and the dielectric caps **132**.

(19) FIG. 6 shows a partially completed structure **100** after removal of the protective liner **130**, according to an embodiment of the disclosure. Referring to FIG. 6, the protective liner **130** may be removed from the silicon fins **148** and **150** and top surfaces of the substrate **110** adjacent to the

silicon fins **148** and **150**. The removal of the protective liner **130** may be by a wet etch or dry etch process. The removal process may leave behind the silicon germanium condensation layer **136** over the silicon fins **152** and **160** and over top surfaces of the substrate **110** adjacent to the silicon fins **152** and **160**. The removal process may also leave behind the dielectric caps **132** over the silicon fins **148**, **150**, **152** and **160**.

(20) FIG. 7 shows a partially completed structure **100** after formation of a dielectric liner **140** and a polysilicon layer **142**, according to an embodiment of the disclosure. Referring to FIG. 7, the dielectric liner **140** may be formed over the silicon germanium condensation layer **136** and the dielectric caps **132** above the silicon fins **152** and **160**, the silicon fins **148** and **150**, the dielectric caps **132** above the silicon fins **148** and **150**, and top surfaces of the substrate **110** adjacent to the silicon fins **148** and **150**. The formation of the dielectric liner **140** may include depositing a layer of a suitable dielectric material, for example, silicon dioxide. The polysilicon layer **142** may be formed over the dielectric liner **140** by depositing a layer of polysilicon. The dielectric liner **140** and the polysilicon layer **142** may be deposited by a suitable deposition process, for example, chemical vapor deposition, atomic layer deposition and physical vapor deposition, or any other suitable deposition processes.

(21) FIG. 8 shows a partially completed structure **100** after formation of an interlayer dielectric layer **122**, according to an embodiment of the disclosure. Referring to FIG. 8, the interlayer dielectric layer **122** may be formed over the polysilicon layer **142**. The formation of the interlayer dielectric layer **122** may include depositing a layer of a suitable dielectric material, for example, silicon dioxide, silicon nitride or a low dielectric constant material. The interlayer dielectric layer **122** may be deposited by a suitable deposition process, for example, chemical vapor deposition, physical vapor deposition, or any other suitable deposition processes.

(22) FIG. 9 shows a partially completed structure **100** after formation of silicon germanium fins **112a** and **112b**, and first **116a**, second **116b** and third **116c** silicon germanium layers, according to an embodiment of the disclosure. Referring to FIG. 9, the formation of the silicon germanium fins **112a** and **112b**, and the first **116a**, second **116b** and third **116c** silicon germanium layers may include a condensation step whereby the structure **100** is annealed in steam at approximately 600° C. for 60 minutes and a higher temperature anneal at approximately 1050° C. for 30 minutes. The condensation step may cause the germanium atoms in the silicon germanium condensation layer **136** to migrate to the silicon fins **152** and **160**, thereby forming the silicon germanium fins **112a** and **112b**, respectively. The condensation step may also cause the germanium atoms in the silicon germanium condensation layer **136** to migrate to upper portions of the substrate **110** near to the first **102a** and second **102b** side surfaces of the silicon germanium fin **112a** and near to the first **106a** and second **106b** side surfaces of the silicon germanium fin **112b**. Although not shown for simplicity, the remaining silicon in the silicon germanium condensation layer **136** may be oxidized to form silicon dioxide, the dielectric liner **140** may remain over the silicon germanium fins **112a** and **112b** and the silicon fins **148** and **150**, and the polysilicon layer **142** may be oxidized to form a silicon dioxide layer.

(23) FIG. 10 shows a partially completed structure **100** after planarization of the interlayer dielectric layer **122** and formation of an opening **146** in the interlayer dielectric layer **122** and the substrate **110**, according to an embodiment of the disclosure. Referring to FIG. 10, a portion of the interlayer dielectric layer **122** may be removed by chemical mechanical polishing during the planarization process. The planarization process may expose top surfaces of the dielectric caps **132**. The top surfaces of the dielectric caps **132** may be co-planar with a top surface of the remaining portion of the interlayer dielectric layer **122** after the planarization process. The dielectric caps **132** protect the top surfaces of the silicon germanium fins **112a** and **112b** and the silicon fins **148** and **150**. The opening **146** may be between the silicon germanium fin **112a** and the silicon fin **150**. The opening **146** may be formed by a suitable photolithography process followed by a wet or dry etch. A portion of the interlayer dielectric layer **122** and the substrate **110** between the silicon germanium

fin **112a** and the silicon fin **150** may be removed to form the opening **146**.

(24) FIG. **11** shows a partially completed structure **100** after formation of an isolation structure **118** and an interlayer dielectric layer **122** between the silicon germanium fin **112a** and the silicon fin **150**, according to an embodiment of the disclosure. Referring to FIG. **11**, the formation of the isolation structure **118** and the interlayer dielectric layer **122** between the silicon germanium fin **112a** and the silicon fin **150** may include depositing a layer of a suitable dielectric material, for example, silicon dioxide or silicon nitride into the opening **146**, thereby filling up the opening **146**. Although not shown, the silicon dioxide layer may also be deposited above the top surface of the interlayer dielectric layer **122** and the dielectric caps **132**. A suitable planarization process, for example, chemical mechanical planarization, may be used to remove a portion of the silicon dioxide layer above the interlayer dielectric layer **122** and the dielectric caps **132**.

(25) The fabrication process may continue to form the structure **100** illustrated in FIG. **1**. Referring to FIG. **1**, a wet etch or dry etch process may be used to remove the dielectric caps **132** from the silicon germanium fins **112a** and **112b** and the silicon fins **148** and **150**, thereby exposing the top surfaces of the silicon germanium fins **112a** and **112b** and the silicon fins **148** and **150**. The etching processes may also remove an upper portion of the interlayer dielectric layer **122**, thereby exposing upper portions of the silicon germanium fins **112a** and **112b** and the silicon fins **148** and **150**. The etching processes may leave behind a lower portion of the interlayer dielectric layer **122** next to lower portions of the silicon germanium fins **112a** and **112b** and the silicon fins **148** and **150**.

(26) The terms “first”, “second”, “third”, and the like in the description and in the claims, if any, are used for distinguishing between similar elements and not necessarily for describing a particular sequential or chronological order. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the device described herein are, for example, capable of operation in sequences other than those illustrated or otherwise described herein. The terms “left”, “right”, “front”, “back”, “top”, “bottom”, “over”, “under”, and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the device described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein. Similarly, if a method is described herein as comprising a series of steps, the order of such steps as presented herein is not necessarily the only order in which such steps may be performed, and certain of the stated steps may possibly be omitted and/or certain other steps not described herein may possibly be added to the method. Furthermore, the terms “comprise”, “include”, “have”, and any variations thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or device that comprises a list of elements is not necessarily limited to those elements, but may include other elements not expressly listed or inherent to such process, method, article, or device.

(27) While several exemplary embodiments have been presented in the above detailed description of the device, it should be appreciated that number of variations exist. It should further be appreciated that the embodiments are only examples, and are not intended to limit the scope, applicability, dimensions, or configuration of the devices in any way. Rather, the above detailed description will provide those skilled in the art with a convenient road map for implementing an exemplary embodiment of the devices, it being understood that various changes may be made in the function and arrangement of elements and method of fabrication described in an exemplary embodiment without departing from the scope of this disclosure as set forth in the appended claims.

Claims

1. A structure comprising: a substrate having a top surface; a first silicon germanium fin over the substrate and extending vertically from the top surface of the substrate, the first silicon germanium fin is of a different material from substrate material of the substrate; a first silicon germanium layer in the substrate and beneath the top surface of the substrate, wherein the first silicon germanium layer contacts the first silicon germanium fin; and a second silicon germanium layer in the substrate and beneath the top surface of the substrate, wherein the second silicon germanium layer contacts the first silicon germanium fin, wherein the first silicon germanium fin has a bottom surface, the first silicon germanium layer has a first side, the second silicon germanium layer has a first side, and the first side of the first silicon germanium layer and the first side of the second silicon germanium layer underlaps the bottom surface of the first silicon germanium fin, and a silicon fin over the substrate; and an isolation structure in the substrate between the first silicon germanium fin and the silicon fin, wherein the first silicon germanium layer has a second side which contacts the isolation structure.
2. The structure of claim 1, wherein the first silicon germanium layer has a top surface, the second silicon germanium layer has a top surface, and the bottom surface of the first silicon germanium fin is coplanar with the top surface of the first silicon germanium layer and the top surface of the second silicon germanium layer.
3. The structure of claim 1, further comprising: a portion of the substrate under the first silicon germanium fin, wherein the portion of the substrate is between the first silicon germanium layer and the second silicon germanium layer.
4. The structure of claim 3, wherein the portion of the substrate has a flat top surface contacting the bottom surface of the first silicon germanium fin.
5. The structure of claim 2, wherein the first silicon germanium layer contacts the second silicon germanium layer.
6. The structure of claim 1, further comprising: a second silicon germanium fin over the substrate, wherein the second silicon germanium fin is spaced from the first silicon germanium fin, and the second silicon germanium layer is coupled to the second silicon germanium fin.
7. The structure of claim 6, wherein the second silicon germanium fin has a bottom surface, the second silicon germanium layer has a second side and the second side of the second silicon germanium layer underlaps the bottom surface of the second silicon germanium fin.
8. The structure of claim 1, wherein the structure is a vertical field effect transistor.
9. The structure of claim 6, wherein the first silicon germanium fin and the second silicon germanium fin have side portions and middle portions, and the side portions of the first silicon germanium fin and the second silicon germanium fin have higher germanium concentration than the middle portions of the first silicon germanium fin and the second silicon germanium fin.
10. The structure of claim 1, wherein the top surface of the substrate is horizontal, and the first silicon germanium layer and the second silicon germanium layer are beneath the horizontal top surface of the substrate, and the first silicon germanium layer has a top surface and the second silicon germanium layer has a top surface, and the top surface of the first silicon germanium layer and the top surface of the second silicon germanium layer are coplanar with the top surface of the substrate.
11. The structure of claim 1, wherein the structure is a vertical nanowire.
12. A structure comprising: a substrate having a top surface; a first silicon germanium fin over the substrate and extending vertically from the top surface of the substrate, the first silicon germanium fin is of a different material from substrate material of the substrate, the first silicon germanium fin having a bottom surface; a first silicon germanium layer in the substrate and beneath the top surface of the substrate, wherein the first silicon germanium layer is coupled to the first silicon germanium fin and the bottom surface of the first silicon germanium fin contacts a portion of the first silicon germanium layer; a second silicon germanium layer in the substrate and beneath the top surface of

the substrate, wherein the second silicon germanium layer is coupled to the first silicon germanium fin and the bottom surface of the first silicon germanium fin contacts a portion of the second silicon germanium layer; and the second silicon germanium layer is spaced from the first silicon germanium layer, wherein the first silicon germanium layer has a first side, the second silicon germanium layer has a first side, and the first side of the first silicon germanium layer and the first side of the second silicon germanium layer underlap the bottom surface of the first silicon germanium fin, and a silicon fin over the substrate; and an isolation structure in the substrate between the first silicon germanium fin and the silicon fin, wherein the first silicon germanium layer has a second side which contacts the isolation structure.

13. The structure of claim 12, wherein a portion of the substrate fully separates the first silicon germanium layer from the second silicon germanium layer.

14. The structure of claim 12, wherein the first silicon germanium layer has a top surface, the second silicon germanium layer has a top surface, and the bottom surface of the first silicon germanium fin is coplanar with the top surface of the first silicon germanium layer and the top surface of the second silicon germanium layer.
